Known Limitations & Assumptions

The Silicon-Power Gap, annex to the master research product Lorenzo Frascolla, Simancas Capital. Updated October 2026 for v39.1 of the master paper.

This annex names every soft spot in the work before a reader can. The thesis direction and order-of-magnitude size are robust to the limitations below; the precision of any individual point estimate is not.

Valuation framework

Black-Scholes-Merton applied to a non-traded real option. The per-slot value of $19.7M is a Trigeorgis-style real-options heuristic (Trigeorgis 1996; Dixit and Pindyck 1994), not a closed-form valuation in the strict sense. The Black-Scholes preconditions of tradeable underlying, continuous delta-hedging, log-normal dynamics, known constant volatility, and no payout leakage (Black and Scholes 1973; Merton 1973; Hull 2021) do not hold for a non-traded, non-replicable transformer slot. Reasonable per-slot bracketing range $15M–$25M; the central illustrative value is $19.7M. The v38 respecification pairs S (the PV of the acceleration increment, discounted, $55.1M) with a strike defined as the carrying cost of pulling the fit-out forward two years ($37.9M) rather than with the full build cost, because S and the old strike were measured on different objects. It also discounts S explicitly; earlier drafts summed two years of scarcity rent undiscounted and called the sum a present value. The thesis depends on the directional sign and the bracketing range, not on the central value.

Volatility input σ = 35% is a proxy. Sourced from H100 GPU spot-price 12-month realized volatility (Lambda Labs 2026; CoreWeave 2025) and applied on the working assumption that transformer-slot economics inherit the volatility of the underlying compute revenue stream. Defensible as a directional anchor in the spirit of practitioner volatility-surface construction (Hull 2021); not defensible as a precision-bearing input. A 5-percentage-point change in σ moves per-slot value by approximately $0.5M (this annex previously said $1.5M; $0.5M is the arithmetically correct figure and the paper now matches it). Full sensitivity surface in the model's `20_Robustness` tab.

Real-options uplift line retired in v7.2. The earlier $115.5M Defer + Expand − Abandon uplift (Trigeorgis 1996; Dixit and Pindyck 1994) was illustrative and its binomial-lattice computation was never independently verified. The v39 bridge is three lines: $315M gross acceleration-option value, −$8M liquidity discount, −$31M thermal liability, netting to $276M. The +$24M correlated-constraint moat premium carried in earlier drafts has been retired: there is no defensible route from a pairwise correlation to a percentage value uplift, the effect belongs in the volatility input rather than as a second additive term, and booking only the slow tail was asymmetric.

DCF uses normalized through-cycle NOPAT. The terminal value (`16_DCF_Terminal`) uses normalized NOPAT inputs rather than projecting from the Income Statement, consistent with the value-driver / through-cycle approach recommended by Koller, Goedhart, and Wessels (2020).

Defensible as a modeling choice for an infrastructure operator; means changes in GPU-hour price do not flow into the DCF via the IS path.

Operator-level estimates

Operator X net hidden value $276M (the $307M figure is the liquidity-adjusted option leg alone). Author's modeled estimate from the v39 three-line bridge: $315M gross acceleration-option value, −$8M liquidity discount, −$31M Sun Belt thermal liability. The +$24M correlated-constraint moat premium and the $348M gross figure both belong to v37 and are retired. Conservative band $260M–$320M (Table A5); wider bracketing range $240M–$400M depending on parameter choices.

Operator H net hidden value $104M (conservative band $95M–$125M). Author's modeled estimate. Bracketing range $95M–$125M (a separate bear-case pass enumerates the impairment scenarios).

Combined two-name book $380M. Author's aggregation; the sum of $276M + $104M. This lands within 5% of the approximately $400M probability-weighted figure from a separate bear-case pass over the same inputs; that is a consistency check, not an independent one. Earlier drafts described the book as falling to $400M, which applied to the retired $476M figure.

Insurance inputs re-based in v39; the geographic adjustment remains an assumption. The transformer insurance base rate is 0.30% of unit replacement value ($27.6K per unit per year), taken from inside a 0.2%–0.5% benchmark band anchored to onshore-wind physical-damage cover (UNEP n.d.), S&P Global Ratings' data-centre premium projection as reported by Insurance Journal (S&P Global Ratings 2026), and the Marsh Q1 2026 commercial-property rate index (Marsh 2026); the $200K per unit per year (2.2%) rate carried through v38 had no market support. The ±40% Sun Belt/Nordic adjustment applied to that base is retained and remains an assumption. Three further Operator H bridge lines are assumptions with no derivations: the + $18M sovereign-siting premium, the −$22M latency penalty, and the +$4M waste-heat line, which recomputes from its stated per-MW rate only under an unstated monetized-MW assumption.

Operator H identity withheld pending diligence. Per the model's Tab 21 note, the specific public-company identity for the Nordic long leg is held against a candidate set (Iron Mountain, Digital Realty, Equinix, Vantage SDC) rather than a specific named company. Any per-share or per-EBITDA figure for Operator H is illustrative against the candidate set.

Input citation status (per 2026-05-16 primary-source check)

Five of the six highest-frequency citations in the model's `01_Inputs` sheet were found on 2026-05-16 to have variance from current primary-source values. The model now carries point-in-time anchor notes in the citation column disclosing this. Full attestations are kept with the working files. Specifically:

- CORRECTED in v39.1: v38 labeled the 4.40% risk-free input as the 1-year UST (FRED DGS1), but that yield was 3.83% on 1 June 2026. The 4.40% is now described as a

conservative input; at 3.83% the per-slot value rises to about $24.1M (valuation model, tab 20).

- Damodaran ERP cited 4.65% vs Jan 2026 update 4.23% (Damodaran 2024).
- ICE BofA BB HY effective yield cited 6.50% (with the wrong FRED ticker BAMLH0A1HYBB, which is the OAS; the correct ticker BAMLH0A1HYBBEY currently reads 5.57%) (ICE BofA 2026; FRED 2026).
- EIA energy $62/MWh: figure is plausible as hyperscaler-contracted PPA but the prior citation to "EIA Form 861 Commercial Average" was incorrect; the actual EIA U.S. commercial average is $143.70/MWh (EIA 2025). Citation now reframed to LBNL DC benchmarks (Shehabi et al. 2024) plus disclosed hyperscaler PPAs.
- Lambda Labs H100 cited $2.45/hr vs Lambda's actual on-demand pricing $2.89/hr (Lambda Labs 2026). Citation now reframed as "H100 spot market median across major providers" (Lambda Labs 2026; CoreWeave 2025).
- Hitachi Energy backlog: the prior $34.6B FY25 figure was the FY25 ORDERS RECEIVED, not the backlog. The actual FY25 backlog per primary is $57.9B (Hitachi Energy 2025). Tear sheet v07 carries the corrected value and the resulting 5x growth multiple (vs the prior 3x).

The cited values were preserved as point-in-time anchors rather than re-pegged to current primaries; the reader is informed of the variance and the model architecture supports re-anchoring cleanly.

Constructed-figure language

The following figures are presented in the prose at a level of confidence the underlying evidence does not fully support. Each should be read as illustrative or as a reasoned estimate within a wider band.

"AI consumes 42% of marginal GOES supply." Author's order-of-magnitude estimate from USGS Mineral Commodity Summaries 2025 GOES production data (USGS 2025) combined with Cushman and Wakefield 2025 data-center transformer demand estimates (Cushman and Wakefield 2025). Defensible range approximately 30–55%; specific 42% figure is the central point.

"Correlation matrix estimated from 2018–2025 quarterly data." The matrix is constructed from a mix of quarterly-resolution market data and lower-frequency industry indicators. The first principal eigenvector explains approximately 76% of the variance under this construction; alternative constructions produce 60–85%. The paper reports "about 80%" and treats the 76% figure as governing. The directional finding (correlated constraints) is robust; the specific eigenvalue is illustrative.

PUE dynamic 1.28 vs static 1.15. Static value is the Uptime Institute 2024 hyperscale median (Uptime Institute 2024). Dynamic value is the author's derivation applying the pump-affinity law (Stepanoff 1957) at 100% utilization under the ASHRAE TC 9.9 thermal envelope (ASHRAE 2021); this is a reasoned estimate, not a directly observed figure.

PUE 1.10 Hamina vs 1.42 Sun Belt. Hamina figure is from Google IR disclosures (Google 2024); Sun Belt figure is the Uptime Institute regional survey median (Uptime Institute 2024). The specific paired comparison is the author's framing.

Free-cooling-hour figures (89% Helsinki / 6% Miami). Author's derivation from FMI/SMHI/NOAA wet-bulb data (Finnish Meteorological Institute 2025; Swedish Meteorological and Hydrological Institute 2025; NOAA 2025) against ASHRAE TC 9.9 thresholds (ASHRAE 2021). Reproducible in `silicon_power_gap_code/05_nordic_comparison.py`. The specific percentages depend on the chosen wet-bulb threshold; ±3 percentage points either way is defensible.

Monte Carlo thermal percentiles in the model's `12_MC_Thermal` tab are typed values calibrated to physical priors rather than simulation output. The simulation engine exists in `silicon_power_gap_code/04_monte_carlo_volatility.py` and should be run with a documented seed and parameter set in a v08 model rebuild.

Engineering corrections made in v38

Cooling class. Earlier drafts declared ONAN/ONAF cooling and then computed an oil pump parasitic load and booked a $41 thousand per unit per year saving from it. Under IEEE C57.12.00 nomenclature the second letter denotes circulation of the internal liquid and N means natural, so ONAN and ONAF have no oil pump; the forced element under ONAF is the radiator fans. The whole cooling auxiliary budget for a 100 MVA unit is on the order of 7 to 8 kilowatts, which cannot support a $41 thousand saving. The pump line has been deleted from the Operator H bridge, worth about $6M of the aggregate.

Loss rate. Earlier drafts took transformer losses at 2.0% of throughput, giving 2.0 MW thermal on a 100 MVA unit. A modern 100 MVA power transformer runs 99.5 to 99.7% efficient: roughly 40 to 60 kW no-load and 250 to 350 kW load loss, about 0.35% of rating. All mass-flow figures have been rerun at 0.35%; the ratios between fluids are unaffected.

Ester fluids. The fluid table previously labelled a column "Pump Parasitic Delta" and populated it with mass-flow deltas, implying natural ester costs about 11% less to circulate. Pump work scales with volumetric flow and viscosity, and natural ester is roughly three times as viscous as mineral oil at operating temperature, so the sign was wrong. The column is relabelled and the ester case is now made on fire safety and on cellulose life extension of roughly five to eight times at equal hot-spot temperature, which is the stronger argument.

Aging baseline. Earlier drafts stated "the IEEE reference ambient of 20 degrees C". IEEE C57.12.00 rates at a 30 degree C average ambient, and the 110 degree C reference hot spot is that ambient plus a 65 degree C average winding rise plus a 15 degree C hot-spot differential. The 20 degree C convention is IEC, which pairs with a 98 degree C reference. On the correct IEEE basis no site in the study ages faster than the laboratory reference, so the thermal section is now built on a relative aging index against the Northern Virginia benchmark rather than on an absolute claim.

Ambient pass-through. Earlier drafts used a 0.594 pass-through from ambient to hot spot and justified it by saying the winding rise dominates. Hot-spot temperature is the additive sum of ambient, top-oil rise and winding gradient, so at constant load the pass-through is one to one and

the justification was a non-sequitur. v38 uses 1:1. The Dallas row previously inverted the sign of its own pass-through and has been recomputed.

Convexity. F_AA is convex in temperature, so evaluating it at an annual mean understates aging, and understates it more at high-variance high latitudes. v38 integrates the aging factor over monthly climate normals. This compresses the Miami-to-Helsinki spread from 8.4x on a mean basis to 6.0x properly integrated, and 6.0x is the figure the paper now uses.

Life reporting. Thermal aging alone implies service lives of 30 to 270 years across the site set, which no fleet observes. v38 reports expected life on a competing-risks basis against a 45-year non-thermal characteristic life, giving 18.6 years at Miami and 36.5 at Helsinki. The financial object is the replacement-capital reserve, not book depreciation, because funds from operations adds depreciation back.

PUE scope. Earlier drafts argued the transformer cooling saving and the facility PUE saving were additive because they cover different equipment. The Green Grid metric puts everything inside the utility meter in the numerator, including transformer losses and their auxiliaries, so they are not additive. The claim has been withdrawn.

Free-cooling threshold. The stated rule omitted the cooling-tower approach and the plate-exchanger approach. v38 states the net threshold and notes that warmer direct-to-chip facility water would raise free-cooling hours everywhere and compress the geographic spread; the conservative legacy threshold is retained, so the Nordic advantage shown is the smaller of the two readings.

Table A2. The final column, previously labelled "Implied sigma (annualized)", is a compound price growth rate and is now labelled as one. A monotone cost curve is evidence of drift, which belongs in S, not of dispersion, which belongs in sigma.

Model structural limitations

Unit economics re-based in v7.2 (formerly a unit error in 05_Unit_Economics). Earlier versions dimensionally treated P_GPUhr as a per-MW-hour price; v7.1 carried tenant-scale compute value (about $20.4M/MW) into the income statement, leaving football-field rows 5 (EV/EBITDA) and 9 (Activist Liquidation) on the wrong revenue basis. v7.2 aligns the unit economics to the paper: gross tenant value $14.2M/MW (η = 0.78, PUE 1.18, ~1,000 accelerator-equivalents per MW at $2.45 per GPU-hour), with the income statement and rows 5 and 9 re-based to landlord lease capture of $2.0M/MW (paper range $1.5M to $2.5M). All six football-field rows now sit on a consistent landlord basis; the BSM / variant-bridge / DCF terminal-value engine was unaffected throughout.

Sources folder partly populated. The `sources/` folder structure is complete and `_index.md` and `_attestations.md` are written, but the primary-source PDFs themselves are not all downloaded locally. A reader who asks “show me the LBNL report” should be given the URL (Shehabi et al. 2024); the local PDF copies are an action item.

Authorship and background

A Note on AI Assistance and Authorship

I used AI tools, primarily Anthropic's Claude, extensively in producing this work. The division of labor was deliberate and consistent throughout.

My role (the brain and orchestrator): I chose the problem and pursued it independently. I directed every line of inquiry. I taught myself the underlying physics, finance, and modeling techniques well enough to evaluate what the AI produced. I made every judgment call on what to keep and what to reject. I verified figures against primary sources where verification was possible, and I documented the variances I found in the *Known Limitations and Assumptions* annex. I take full responsibility for every claim in this document.

The AI's role (the hands and facilitator): It generated initial framings I then refined or discarded. It drafted the prose, which I directed and edited. It suggested modeling structures I then implemented and stress-tested. It assembled first-pass citation candidates I then verified against primary sources where possible. It executed routine production tasks, document formatting, table layout, cross-reference checks, under my direction.

The substance is collaboratively produced. The accountability is mine alone. I am a rising senior (through August 2026) at Gulliver Preparatory School (IB Diploma May 2027: Physics HL, Mathematics Analysis and Approaches Standard Level) (International Baccalaureate Organization 2023a, 2023b). Mathematical techniques used beyond the IB Standard Level curriculum, Weibull distributions, Black-Scholes-Merton option pricing, principal component analysis, and Monte Carlo simulation, were taught to me from open-access sources, primarily MIT OpenCourseWare and the textbooks listed in the Bibliography.

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